

Platforms / Simulation: Isaac Sim / Isaac Lab / Isaac ROS, Jetson Thor / Orin, PyTorch / MuJoCo

Control / Engineering: RL / Sim2Real / Whole-Body Control, ROS 2, C++ / Python

Education

2024.09–Present	Beihang University (BUAA)	Cyber Science and Technology	M.S.
2020.09–2024.06	Southwest University	Computer and Information Science	B.Eng.

Experience

2025.11–Present

National Key R&D Program · B2 Quadruped-Manipulator Inspection Robot

Perception & Control Lead

- Led sim-to-real delivery of an **18-DoF whole-body RL controller**, spanning Isaac Lab training, actuator modeling, model packaging, a C++ real-time stack, and Nav2 integration.
- Policy training:** built a **12 leg DoFs + 6 arm DoFs** Isaac Lab / PPO environment with observation history and privileged adaptation. With no end-effector force sensor, recast force–position control as coordinated base-velocity and end-effector-position tracking while preserving the pretrained observation contract.
 - Actuator ID & Sim2Sim:** identified per-joint PD gains / delays and used Sim2Sim to rebuild training-side actuators as **explicit PD models**: delayed PD for Unitree leg-CAN latency, and first-order low-pass + delayed PD for the arm. Derived rate-equivalent filters across Isaac Lab (200 Hz), MuJoCo, and hardware (1 kHz); raised arm stiffness for position-servo behavior, synced the parameters into training, and randomized friction / CoM / payload, eliminating arm oscillation while preserving stable locomotion.
 - Deployment:** defined a **SHA-verified ONNX + config bundle** and shared FSM across sim and hardware. Built a **50 Hz inference / 1 kHz DDS-CAN C++ runtime** with safety checks, communication probes, and dual-rate logs; resolved joint-order, gait-phase, and zero-fallback mismatches.
 - Drift correction:** added a FAST-LIVO2 command-space PI outer loop to the frozen policy; **without retraining**, cut 25 m lateral drift from 5.18 m to 0.15 m (**97%**), meeting Nav2 and manipulation accuracy requirements.

2026.06–Present

Tsinghua Qiyuan Lab · Robot-Dog Swarm Control & Digital Twin

Isaac ROS Infrastructure Lead

- Own low-level control, GPU perception, and ROS 2 interfaces for B2 + R5A (**Isaac Sim 5.1 / Isaac ROS 4.5 / Jetson AGX Thor**), supporting VLN, segmentation, and VLA research.
- Isaac ROS bring-up:** brought up the full Isaac ROS 4.5 stack on Thor; wrote a **ros2_control plugin** and URDF / SRDF / XRDF for the unsupported R5A, wrapping vendor CAN so MoveIt 2 and cuMotion drive hardware **without modifying NVIDIA source**.
 - Task orchestration:** standardized perception outputs as target-pose / scene-obstacle interfaces to a custom **cuMotion pipeline**; tuned **Isaac ROS Manipulation behavior-tree** nodes so low-rate high-level commands trigger local planning and closed-loop execution for door opening and debris clearing. A **PC–Thor DDS** setup reuses interfaces across sim and hardware.
 - Rapid deployment:** deployed LocateAnything-3B **on Thor** to replace the YOLO front end and integrated fine-tuned **DINOv3** segmentation, enabling same-day model-to-robot validation.

2026.05–2026.07

DongYi Tech · Phybot C2 Humanoid Soccer & Sim2Real

RL & Sim2Real Engineer

- Delivered one **Isaac Lab–MuJoCo–hardware** path for four skills on a **21-DoF humanoid**: walking, penalty kick, soccer kick, and drop kick.
- Multi-skill policies:** trained walking and kicking with **AMP / PPO**; unified walking / penalty kick and added reference-motion residual control plus dedicated FSMs for dynamic kicks.
 - Contract alignment:** matched joint order, 5-frame history, action scale, PD gains, and ball semantics across training and C++; separated yaw-free gravity projection from yaw-aware heading alignment, eliminating observation-induced hardware failures.
 - Real-time deployment:** built a shared TorchScript path (**500 Hz motor control / 50 Hz inference**) with zero calibration, safety FSM, and aligned logs; validated all four skills on hardware.

2025.03–2026.04

China Aerospace 9th Academy · 30-UAV Swarm

Swarm Planning & Onboard Software

- Owned Jetson computing, mission planning, and communications for a **30-UAV swarm**; led the ROS 1-to-ROS 2 migration and field validation.
- Onboard planning:** integrated PX4, MAVROS, and visual / inertial sensors on **Orin NX**; implemented 3D space-time weighted A* with conflict resolution and used **Zenoh** for cross-subnet discovery / bandwidth, validating **public-5G swarm control** in field tests.
 - Software infrastructure:** built an Orin NX **robot app manager** (FastAPI / Vue 3 / systemd) for install, versioning, lifecycle, boot, and cloud distribution; delivered a **publish–distribute–one-click-deploy** loop and unified fleet telemetry through Foxglove + Zenoh.

2024.11–2025.10

Beijing Zhuoyi Tech · Hangfeng Autonomous UAV

Deputy Chief Designer

- Owned navigation-stack integration and vehicle interfaces from system design through prototype flight tests.
- Integrated **FAST-LIO2 / EGO-Planner / RACER**; unified timing, frames, calibration, and state-machine interfaces, and built log-based regression to diagnose drift, oscillation, and planning timeouts during field and formation tests.

Publications

Title	Venue	Role	Status
Odometry-Guided Command-Space Drift Correction for a Frozen RL Quadruped-Arm Policy	ICARCV 2026	First author	Submitted
Security-Enhanced Formation Tracking Control for Networked Swarm Systems via Quantized Encryption-Decryption	IEEE TCNS	Co-first author	Accepted
FormationDiffuser: Boundary-Consistent Diffusion for Long-Horizon 3D Multi-UAV Reconfiguration	AAAI 2027	Co-first author	Submitted
IRA: Interception-Risk-Aware Trajectory Planning for MAV Evasion from Multiple Pursuers in Cluttered Environments	IEEE RA-L	Second author	Under Review